Supplementary material
Separating task information from evaluation effects in language-model performance

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This supplement gives the complete result tables, analytical definitions, execution details, and audit scope for the main paper. The two empirical studies are evaluated separately. Their combination is an interpretation of the evidence, not a joint test of a fitted capability theory.

S1 Analysis status and retained development evidence

The main paper is organised around the results rather than around the order in which the analyses were developed. This presentation does not change the evidential status of any comparison. The studies originated in a programme distinguishing capability, task demands, and execution conditions. No universal transformation model, causal decomposition of intelligence, or uniquely correct demand taxonomy is established here.

For Study 1, the original response-conditioned comparisons used leave-one-task-family-out validation. A later extension fixed six predictive specifications and introduced whole-benchmark holdout. Its primary contrast was R versus $R+T_{768}$ under fixed $C = 1$. That comparison did not show a gain. All specifications containing D were designated descriptive analyses of an existing representation. Their stable positive increment is reported without relabelling it a successful primary endpoint.

The regularisation grid $\{0.01, 0.1, 1, 10\}$ was fixed with the transfer extension. The exact three-fold, benchmark-grouped inner validation scheme was documented after the fixed-C outputs existed and before the nested sensitivity ran. It is a transparent implementation decision, not a detail independently preregistered before any results. The observation that all 300 full-embedding fits selected the lower grid boundary motivated a later post hoc sensitivity extending the grid to $\{0.0001, 0.001, 0.01, 0.1, 1, 10\}$. Direct representation intervals reported in Section S4 were also computed after the transfer outputs became available.

For Study 2, inspection of 24 development tasks revealed missing output wrappers in Gemma's natural-language responses. Mechanical wrapping of those outputs was an outcome-aware sensitivity. The four-arm protocol, tolerant extraction rule, and primary contrasts were then fixed before generating responses on the 72 reserved tasks. Primary inference uses two exact paired tests per model with $\alpha = 0.025$. Wrapper and compilation endpoints were recorded separately. The score-accounting identity and task-source split are explanatory analyses of the retained outputs, not additional prospectively specified primary tests.

The development programme also tested richer predictors of programming outcomes. A compact model containing model, input condition, their interaction, task source, and difficulty had task-held-out log loss 0.57375. Adding five task proxies and model interactions gave 0.65025. Further environment interactions gave 0.67364. A duplicated difficulty column altered the L2 penalty in the richer designs. Removing it changed the latter losses to 0.64892 and 0.67218, leaving the original predictive gate unmet. The gate-dependent confirmation was not run. The later 72-task measurement experiment answered a separately specified question and does not fulfil that earlier gate.

Local protocols and hashes record the retained chronology. They establish content identity relative to those records, not independent certification of the time at which an analysis was conceived. This manuscript does not describe local freezes as external preregistrations.

S2 Study 1 data and estimator

The data are the released ADeLe battery and fifteen subject response files (Zhou et al., 2026). Items must have finite binary `verification_final` values for all targets, complete annotations, and UG $\geq$ 75. All results use the same complete-case set of 16,108 items. There are 63 recorded task families and twenty benchmarks. Prediction is performed separately for each of the fifteen targets.

The target file identifiers are `DK-R1-Dist-Qwen-1.5B`, `DK-R1-Dist-Qwen-7B`, `DK-R1-Dist-Qwen-14B`, `DK-R1-Dist-Qwen-32B`, `babbage-002`, `davinci-002`, `gpt-35-turbo`, `gpt4o`, `llama3d1-405b`, `llama3d2-1b`, `llama3d2-3b`, `llama3d2-11b`, `llama3d2-90b`, `o1-mini`, and `o1_re=low`. These are dataset identifiers rather than claims about current model offerings.

For target m and item i, let $r_{i,-m}$ contain fourteen peer outcomes. The scalar peer summary is

$$q_{i,-m} = \frac{0.5 + \sum_{\ell \neq m} y_{\ell i}}{15}, \qquad s_{i,-m} = \log \frac{q_{i,-m}}{1 - q_{i,-m}}. \tag{S1}$$

The baseline linear predictor is

$$\eta^R_{mi} = a_m + b_m s_{i,-m} + u_m^\top r_{i,-m} + v_m^\top h_i, \qquad p^R_{mi} = \text{logistic}(\eta^R_{mi}), \tag{S2}$$

where h_i is benchmark identity. Every augmentation refits this baseline together with its additional features. No target outcome is used in that target's peer-response vector.

The eighteen canonical fields in D are `AS`, `CEc`, `CEe`, `CL`, `KNn`, `KNa`, `KNc`, `KNf`, `KNs`, `MCt`, `MCu`, `MCr`, `MS`, `QLq`, `QLl`, `SNs`, `VO`, and `AT`. The published rubrics supply their meanings. `UG` is not a predictive feature.

The frozen Nomic encoder has 768 output dimensions, 137 million parameters, F16 weights, and local model identifier `0a109f422b47`. Requests use only the canonical `question` text, prefixed by `classification:`. Truncation is enabled. Its frequency was not logged. For T_{18}, PCA uses the randomized solver, eighteen components, no whitening, and seed 20260926. PCA is followed by numerical standardisation. T_{768} uses the complete standardised embedding without PCA.

Numerical standardisation and benchmark encoding are fitted on training rows. The benchmark encoder ignores unknown categories. Every held-out benchmark is verified absent from the outer training set and maps to a zero indicator block. The fitted classifier is L2 logistic regression with `liblinear`, maximum 4,000 iterations, and seed 20260926. Fixed-C analyses use $C = 1$.

Nested selection fits every candidate C in each of three inner `GroupKFold` splits grouped by benchmark. The main nested sensitivity uses $C \in \{0.01, 0.1, 1, 10\}$. The post hoc extended-grid sensitivity uses $C \in \{0.0001, 0.001, 0.01, 0.1, 1, 10\}$ with the same inner splits and selection rule. The selection criterion is pooled inner log loss, not the mean of benchmark-specific losses. Each representation and target selects its own C. A tie selects the smaller value. All preprocessing is refitted inside the inner split, then fitted again on the complete outer training set for evaluation on the excluded benchmark. This gives 300 outer fits per representation after selection.

S3 Study 1 results and benchmark dependence

Table S1 reports all predictive metrics. Table S2 reports every augmentation against the corresponding R baseline. Tables S3 to S5 show the twenty benchmark-specific gains under each analysis. The task-family and benchmark evaluations reuse the same items and model outcomes. They are different validation splits, not independent replications.

The primary score is binary log loss, with probabilities clipped to $[10^{-6}, 1-10^{-6}]$. The secondary Brier score is the mean squared error of the predicted probabilities. Accuracy uses a threshold of 0.5 and is supplemental. Log loss and Brier assess probabilistic prediction rather than generated-program success (Gneiting & Raftery, 2007).

For compared predictors A and B, define $d_j = \ell(y_j, p_{Aj}) - \ell(y_j, p_{Bj})$. Within benchmark g, retain its paired sum S_g and row count n_g. Each of 10,000 bootstrap draws samples twenty benchmark labels with replacement and computes

$$\Delta^* = \frac{\sum_{b=1}^{20} S_{g_b^*}}{\sum_{b=1}^{20} n_{g_b^*}}. \tag{S3}$$

Reported intervals are the 2.5th and 97.5th percentiles. The source calculation uses seed 20260926 and a continuous random-number stream through the five augmentation contrasts in table order. All targets within a benchmark remain grouped. Equal-benchmark means are also shown so that large benchmarks do not determine that summary.

These intervals condition on saved predictions and the observed target panel. They do not refit the complete cross-validation procedure, account for all uncertainty from overlapping training sets, or guarantee coverage for a new population of models or benchmarks (Bates et al., 2024). Benchmark sign counts describe direction. They are not twenty independent significance tests.

Table S1: All predictive metrics. Accuracy uses threshold 0.5 and is supplemental. All columns describe the external predictor, not a change in candidate-model success.

Setting	Predictor	Log loss	Brier	Accuracy (%)
Task-family, fixed C	R	0.35665	0.11254	84.14
	$R + D$	0.35045	0.11038	84.45
	$R + T_{18}$	0.35144	0.11080	84.32
	$R + T_{18} + D$	0.34985	0.11024	84.44
	$R + T_{768}$	0.39325	0.12323	82.64
	$R + T_{768} + D$	0.39271	0.12292	82.69
Benchmark, fixed C	R	0.37402	0.11784	83.42
	$R + D$	0.36775	0.11615	83.57
	$R + T_{18}$	0.37203	0.11726	83.43
	$R + T_{18} + D$	0.36963	0.11677	83.42
	$R + T_{768}$	0.45564	0.13947	80.69
	$R + T_{768} + D$	0.45751	0.13983	80.61
Benchmark, nested C	R	0.37253	0.11728	83.47
	$R + D$	0.36627	0.11553	83.70
	$R + T_{18}$	0.36831	0.11604	83.61
	$R + T_{18} + D$	0.36660	0.11574	83.53
	$R + T_{768}$	0.38647	0.12256	82.58
	$R + T_{768} + D$	0.38717	0.12316	82.43

Table S2: Paired gains over R. Positive differences favour the augmented predictor. Intervals are 10,000-draw benchmark-cluster percentiles conditional on saved predictions.

Setting	Predictor	Gain	95% interval	Equal-benchmark	Positive
Task-family, fixed C	$R + D$	+0.00620	[+0.00009, +0.01066]	+0.00589	17/20
	$R + T_{18}$	+0.00521	[+0.00068, +0.00886]	+0.00458	14/20
	$R + T_{18} + D$	+0.00681	[+0.00113, +0.01123]	+0.00648	16/20
	$R + T_{768}$	-0.03660	[-0.04902, -0.02558]	-0.06248	2/20
	$R + T_{768} + D$	-0.03605	[-0.04839, -0.02552]	-0.05979	2/20
Benchmark, fixed C	$R + D$	+0.00628	[+0.00204, +0.01100]	+0.00723	16/20
	$R + T_{18}$	+0.00200	[-0.00158, +0.00655]	+0.00417	14/20
	$R + T_{18} + D$	+0.00439	[+0.00080, +0.01078]	+0.00654	14/20
	$R + T_{768}$	-0.08162	[-0.10942, -0.04068]	-0.08117	0/20
	$R + T_{768} + D$	-0.08349	[-0.11265, -0.04324]	-0.08112	0/20
Benchmark, nested C	$R + D$	+0.00625	[+0.00126, +0.01122]	+0.00707	16/20
	$R + T_{18}$	+0.00421	[-0.00034, +0.00968]	+0.00525	14/20
	$R + T_{18} + D$	+0.00593	[+0.00104, +0.01194]	+0.00716	15/20
	$R + T_{768}$	-0.01395	[-0.02175, -0.00131]	-0.01614	6/20
	$R + T_{768} + D$	-0.01465	[-0.02266, -0.00278]	-0.01604	6/20

Table S3: Task-family, fixed C. Per-benchmark gains over R, averaged over the fifteen targets. No benchmark is omitted.

Benchmark	Items	D	T_{18}	$T_{18} + D$	T_{768}	$T_{768} + D$
ChemLLMBench	1723	-0.0048	+0.0027	-0.0025	-0.0127	-0.0167
Civil Service Examination	408	+0.0080	+0.0040	+0.0059	-0.0637	-0.0643
Data Analysis	33	+0.0086	+0.0040	+0.0066	-0.2750	-0.2745
Date Arithmetic	493	-0.0111	+0.0018	-0.0086	-0.0698	-0.0789
GRE & GMAT	203	+0.0198	+0.0170	+0.0236	-0.0038	-0.0007
LSAT	910	+0.0073	+0.0051	+0.0079	-0.0569	-0.0584
Language	29	+0.0018	-0.0071	-0.0076	-0.1426	-0.1350
MCTACO	205	+0.0059	-0.0019	-0.0002	-0.0656	-0.0663
MMLU-Pro	5431	+0.0123	+0.0092	+0.0123	-0.0368	-0.0354
Math	173	-0.0064	+0.0079	+0.0007	-0.0855	-0.0915
MedCalcBench	556	+0.0023	+0.0020	+0.0051	-0.0651	-0.0593
MenatQA	680	+0.0008	-0.0005	+0.0008	-0.0189	-0.0187
OmniMath	1664	+0.0012	-0.0032	+0.0001	-0.0459	-0.0446
Reasoning	56	+0.0076	-0.0012	+0.0140	-0.1093	-0.0673
SAT	410	+0.0521	+0.0489	+0.0551	+0.0008	+0.0053
SciBench	355	+0.0044	-0.0016	+0.0048	-0.0283	-0.0210
TempReason	657	+0.0027	+0.0021	+0.0027	-0.0189	-0.0194
TimeDial	340	+0.0018	+0.0013	+0.0033	-0.1298	-0.1299
TimeQA	727	+0.0026	+0.0001	+0.0024	-0.0234	-0.0216
TruthQuest	1055	+0.0011	+0.0012	+0.0033	+0.0016	+0.0025

Table S4: Benchmark, fixed C. Per-benchmark gains over R, averaged over the fifteen targets. No benchmark is omitted.

Benchmark	Items	D	T_{18}	$T_{18}+D$	T_{768}	$T_{768}+D$
ChemLLMBench	1723	-0.0012	-0.0087	+0.0005	-0.0257	-0.0333
Civil Service Examination	408	+0.0080	+0.0040	+0.0059	-0.0637	-0.0643
Data Analysis	33	+0.0086	+0.0040	+0.0066	-0.2750	-0.2745
Date Arithmetic	493	-0.0111	+0.0018	-0.0086	-0.0698	-0.0789
GRE & GMAT	203	+0.0198	+0.0170	+0.0236	-0.0038	-0.0007
LSAT	910	+0.0085	-0.0019	+0.0034	-0.0731	-0.0729
Language	29	+0.0018	-0.0071	-0.0076	-0.1426	-0.1350
MCTACO	205	+0.0059	-0.0019	-0.0002	-0.0656	-0.0663
MMLU-Pro	5431	+0.0057	+0.0008	-0.0003	-0.1306	-0.1354
Math	173	+0.0146	+0.0183	+0.0143	-0.0669	-0.0740
MedCalcBench	556	-0.0029	+0.0076	+0.0053	-0.1400	-0.1572
MenatQA	680	+0.0075	+0.0003	-0.0007	-0.1138	-0.1128
OmniMath	1664	+0.0177	+0.0086	+0.0163	-0.0799	-0.0661
Reasoning	56	+0.0080	-0.0039	+0.0024	-0.1420	-0.1256
SAT	410	+0.0117	+0.0116	+0.0167	-0.0401	-0.0351
SciBench	355	+0.0166	+0.0183	+0.0224	-0.0098	-0.0032
TempReason	657	+0.0121	+0.0107	+0.0122	-0.0386	-0.0456
TimeDial	340	+0.0018	+0.0013	+0.0033	-0.1298	-0.1299
TimeQA	727	+0.0199	+0.0125	+0.0211	-0.0102	-0.0070
TruthQuest	1055	-0.0082	-0.0099	-0.0055	-0.0023	-0.0048

Table S5: Benchmark, nested C. Per-benchmark gains over R, averaged over the fifteen targets. No benchmark is omitted.

Benchmark	Items	D	T_{18}	$T_{18}+D$	T_{768}	$T_{768}+D$
ChemLLMBench	1723	-0.0043	-0.0044	-0.0039	-0.0037	-0.0097
Civil Service Examination	408	+0.0054	+0.0029	+0.0049	-0.0145	-0.0141
Data Analysis	33	+0.0090	+0.0014	+0.0072	-0.0585	-0.0584
Date Arithmetic	493	-0.0123	-0.0018	-0.0083	-0.0219	-0.0284
GRE & GMAT	203	+0.0204	+0.0149	+0.0224	+0.0016	+0.0068
LSAT	910	+0.0048	-0.0007	+0.0009	-0.0204	-0.0196
Language	29	-0.0019	-0.0140	-0.0143	-0.0670	-0.0634
MCTACO	205	+0.0111	+0.0069	+0.0083	-0.0102	-0.0120
MMLU-Pro	5431	+0.0066	+0.0031	+0.0048	-0.0253	-0.0261
Math	173	+0.0153	+0.0219	+0.0193	+0.0023	+0.0005
MedCalcBench	556	+0.0025	+0.0112	+0.0119	-0.0532	-0.0630
MenatQA	680	+0.0109	+0.0094	+0.0055	-0.0189	-0.0191
OmniMath	1664	+0.0198	+0.0135	+0.0163	-0.0106	-0.0013
Reasoning	56	+0.0053	-0.0058	-0.0018	-0.0294	-0.0229
SAT	410	+0.0120	+0.0101	+0.0166	+0.0048	+0.0079
SciBench	355	+0.0171	+0.0171	+0.0205	+0.0037	+0.0108
TempReason	657	+0.0117	+0.0150	+0.0169	-0.0083	-0.0112
TimeDial	340	+0.0028	+0.0026	+0.0041	-0.0341	-0.0315
TimeQA	727	+0.0143	+0.0189	+0.0262	+0.0238	+0.0254
TruthQuest	1055	-0.0092	-0.0171	-0.0143	+0.0172	+0.0085

S4 Representation comparisons and regularisation

The direct comparisons in Table S6 address a different question from whether either representation improves on R. An interval above zero for R minus $R+D$ and an interval spanning zero for R minus $R+T_{18}$ do

not themselves show that D outperforms T_{18}. The appropriate statistic is their paired loss difference on the same observations.

At fixed C, $R + D$ has lower loss than $R + T_{18}$ by 0.00428, with a benchmark-cluster interval above zero. Under nested selection that difference is 0.00204 and its interval spans zero. The stable result is the positive annotation increment over R. A superiority claim over compact text is less stable across the tested estimation choices.

Combining the compact text representation with D does not clearly improve on D alone under benchmark holdout. The nested conditional addition of D after T_{18} is positive, but these post-result direct intervals are not multiplicity-adjusted. They are reported together rather than selected by their significance labels. Their bootstrap uses the same paired benchmark procedure with seed 20260927 for each direct contrast.

Table S7 records every penalty selected in the original nested grid. All full-embedding fits choose $C = 0.01$, the strongest value then offered. Tuning reduces the full embedding's excess benchmark-held-out loss from 0.08162 to 0.01395, but the boundary selection leaves open whether still stronger penalisation would alter that conclusion.

A post hoc extended-grid sensitivity therefore adds $C = 0.001$ and $C = 0.0001$ without changing the folds, preprocessing, or inner selection criterion. Table S8 reports the result. The baseline loss remains 0.37253. Full-embedding loss is 0.38687, an excess of 0.01434 or 3.85%. The paired benchmark-cluster interval for $L(R) - L(R + T_{768})$ is $[-0.02136, -0.00197]$, with positive differences on three benchmarks and negative differences on seventeen. The new lower boundary is not selected. Among the 300 full-embedding fits, 104 choose $C = 0.001$ and 196 choose $C = 0.01$. For $R + T_{768} + D$, 115 choose $C = 0.001$ and 185 choose $C = 0.01$. None choose $C = 0.0001$. The original primary comparison remains the frozen fixed-C analysis. The extended grid is a later fairness check on the high-dimensional comparator.

This does not establish that the embedding contains no useful information. Different scaling, encoders, estimators, or still broader modelling choices could produce a different result. It does show that the observed deficit is not explained by stopping at the original $C = 0.01$ boundary.

Table S6: Direct loss contrasts, calculated after the transfer outputs were available. A positive value favours the right-hand predictor. These intervals are unadjusted for multiple comparisons.

Setting	Left minus right	Difference	95% interval
Benchmark, fixed C	$R + T_{18}$ minus $R + D$	+0.00428	[+0.00113, +0.00624]
	$R + T_{18}$ minus $R + T_{18} + D$	+0.00239	[-0.00008, +0.00588]
	$R + D$ minus $R + T_{18} + D$	-0.00189	[-0.00420, +0.00169]
	$R + T_{768}$ minus $R + D$	+0.08789	[+0.04533, +0.11601]
Benchmark, nested C	$R + T_{18}$ minus $R + D$	+0.00204	[-0.00102, +0.00376]
	$R + T_{18}$ minus $R + T_{18} + D$	+0.00172	[+0.00052, +0.00285]
	$R + D$ minus $R + T_{18} + D$	-0.00033	[-0.00194, +0.00268]
	$R + T_{768}$ minus $R + D$	+0.02020	[+0.00543, +0.02902]

Table S7: Selected regularisation across 300 target-by-benchmark fits per representation. Smaller C means stronger penalisation.

Predictor	$C = 0.01$	$C = 0.1$	$C = 1$	$C = 10$
R	39	167	35	59
$R + D$	50	160	40	50
$R + T_{18}$	79	196	18	7
$R + T_{18} + D$	100	160	31	9
$R + T_{768}$	300	0	0	0
$R + T_{768} + D$	300	0	0	0

Table S8: Post-hoc extended-grid sensitivity. Selected regularisation across 300 target-by-benchmark fits per representation. Smaller C means stronger penalisation.

Predictor	$C = 0.0001$	$C = 0.001$	$C = 0.01$	$C = 0.1$	$C = 1$	$C = 10$
R	0	0	39	167	35	59
$R + D$	0	0	50	160	40	50
$R + T_{18}$	0	0	79	196	18	7
$R + T_{18} + D$	0	0	100	160	31	9
$R + T_{768}$	0	104	196	0	0	0
$R + T_{768} + D$	0	115	185	0	0	0

S5 Programming tasks, prompts, and evaluation

The material comes from VeriContest (Xie et al., 2026). A deterministic split within public difficulty strata assigned 24 tasks to development and 72 to the reserve. Development contains six Codeforces and eighteen LeetCode tasks. The reserve contains eighteen Codeforces and 54 LeetCode tasks. The full 96-task substrate passed a technical preflight before candidate execution. Selection was not a random sample of all programming work.

The two retained checkpoints are Devstral Small 2 24B and Gemma 3 12B, both local Q4_K_M builds. Their recorded Ollama identifiers are `24277f07f62d` and `f4031aab637d`. A six-task development screen required at least fifteen extracted implementations, twelve compilations, and between one and seventeen test successes across eighteen attempts per model. Both retained models passed. GPT-OSS 20B achieved eighteen successes and was excluded by that screen's ceiling rule. The reserve is independent of development outcomes at protocol freeze, but the model panel is conditional on this selection.

Each model attempts every reserved task once under NL, NL+scaffold, SPEC, and BOTH. NL contains the natural-language description before its Starter Code section. NL+scaffold adds a `Solution` declaration and an `impl Solution` block with the target signature and an empty body. SPEC supplies the original Verus specification. BOTH combines the NL description and that unchanged specification. The target stable-Rust signature and common instruction are included in all arms. Specification-containing prompts may expose constraints, helper definitions, constructive information, and additional tokens. They are not a pure manipulation of notation.

The instruction requests a single `impl Solution` block using stable Rust 2021 and the standard library, with no explanations, proofs, or Verus annotations. It forbids a separate `main` or `struct Solution`, preserves the target name and types, and permits private helper methods inside the wrapper. No compiler feedback or model repair round is provided. The runner uses Ollama 0.34.0, temperature zero, seed 20260926, context capacity 32,768, and a maximum of 4,096 generated tokens. Requests do not pass a conversation history between attempts. Execution is ordered by model, task, and arm, with the arm order NL, NL+scaffold, SPEC, BOTH. Order was not randomised. Repeated runs did not measure numerical or backend nondeterminism.

The original extractor accepts a balanced `impl Solution` block, including one inside a Markdown code fence. Thus "wrapper compliance" means extraction of the requested structure, not compliance with every formatting instruction. The tolerant rule first runs that extractor. If it fails, the rule removes an enclosing whole-response Rust fence where present. A response beginning with a bare `pub fn` and containing no `impl Solution` is enclosed in the wrapper. The generated function text is unchanged. Other extraction failures remain failures. Name, type, scope, and implementation errors are still exposed by compilation or execution.

Accepted code is spliced into the task's fixed harness and compiled using Rust 1.98.1, edition 2021, optimisation enabled, and a 120-second compilation timeout. Codeforces tests run separately with a ten-second execution limit per case and normalised trailing output whitespace. LeetCode tests run in batches of up to 200 with a sixty-second limit per batch, comparing parsed JSON outputs. These procedures are fixed across input arms within each task. Their difference between sources is one reason not to assign an intrinsic task-demand interpretation to the exploratory source split.

A test success requires compilation and correct output on every packaged positive case. Extraction failures, compilation errors, wrong outputs, runtime failures, and timeouts count as failures. The tests are public benchmark resources withheld from candidate prompts, not newly concealed tests. Passing them is the operational endpoint, not formal verification of the program.

S6 Programming contrasts and score accounting

Table S9 reports all prespecified within-model contrasts. For each paired contrast, b is the number of control failures becoming treatment successes and c is the reverse. The point effect is $100(b-c)/72$ percentage points. The exact two-sided conditional McNemar calculation (Fagerland et al., 2013) uses a binomial distribution with $b+c$ trials and probability one half. When no discordant pairs exist, the value is one. Primary decisions use $\alpha = 0.025$ for each of two tests within a model. This controls that within-model two-test family, not one four-test family pooled across both models.

Table S9: All prespecified within-model semantic-test contrasts. Secondary values are unadjusted. Primary decisions use $\alpha = 0.025$ within each model.

Model	Treatment minus control	Status	Effect (pp)	Gains/losses	Exact p
Devstral	NL+scaffold minus NL	Primary	+2.8	7/5	0.7744
Devstral	BOTH minus NL+scaffold	Primary	+6.9	14/9	0.4049
Devstral	SPEC minus NL+scaffold	Secondary	-9.7	11/18	0.2649
Devstral	SPEC minus NL	Secondary	-6.9	10/15	0.4244
Devstral	BOTH minus NL	Secondary	+9.7	15/8	0.2100
Gemma	NL+scaffold minus NL	Primary	+1.4	3/2	1.0000
Gemma	BOTH minus NL+scaffold	Primary	+9.7	9/2	0.0654
Gemma	SPEC minus NL+scaffold	Secondary	+1.4	13/12	1.0000
Gemma	SPEC minus NL	Secondary	+2.8	14/12	0.8450
Gemma	BOTH minus NL	Secondary	+11.1	10/2	0.0386

Between-model effect differences are secondary. Their task-level values subtract Devstral's arm contrast from Gemma's. These differences can take values $-2, -1, 0, 1, 2$. The specified exact sign test compares positive with negative differences after omitting zeros. It tests the balance of directions, not directly a null hypothesis about the mean difference in percentage points. Table S10 therefore reports both the mean effect difference and the sign-test result without treating them as the same statistic.

Table S10: All prespecified between-model differences. The direction is Gemma minus Devstral. Results are secondary and unadjusted.

Arm contrast	Mean difference (pp)	Positive/negative	Exact sign p
NL+scaffold minus NL	−1.4	8/9	1.0000
BOTH minus NL+scaffold	+2.8	14/13	1.0000
SPEC minus NL+scaffold	+11.1	23/15	0.2559
SPEC minus NL	+9.7	21/14	0.3105
BOTH minus NL	+1.4	14/13	1.0000

Table S11: Execution diagnostics for all 576 confirmation cells. Each row has 72 attempts. Timeout means program execution exceeded the applicable time limit. Length means generation reached its 4,096-token limit.

Model	Input	Wrapper	Wrapped	Compiles	Passes	Timeout	Length
Devstral	NL	72	0	64	39	1	0
Devstral	NL+scaffold	72	0	66	41	1	0
Devstral	SPEC	71	0	60	34	0	1
Devstral	BOTH	72	0	63	46	0	0
Gemma	NL	36	36	55	28	5	1
Gemma	NL+scaffold	72	0	61	29	5	0
Gemma	SPEC	72	0	62	30	3	0
Gemma	BOTH	72	0	65	36	7	0

The strict endpoint is reconstructed as tolerant test success together with successful original wrapper extraction. For Gemma, NL gives twelve strict successes and 28 tolerant successes. SPEC gives thirty under both rules. Consequently the SPEC-minus-NL contrast is $18/72 = 25.0$ points under strict scoring and $2/72 = 2.8$ points under tolerant scoring. Their difference is $16/72 = 22.2$ points. Under strict scoring the discordant counts are 22 gains and four losses, giving $p = 0.0005335$. Under tolerant scoring they are fourteen gains and twelve losses, giving $p = 0.8450$. The generators and test cases are identical in this comparison.

Sixteen of the 36 bare NL responses pass after wrapping. Twenty-six compile. Among the 36 originally extracted NL responses, twelve pass and 29 compile. These are descriptive subsets selected by generated output form. Their comparison is not a randomised test of whether omission predicts better solutions.

Sixteen of the eighteen successes in the strict SPEC-minus-NL gap are recovered by the tolerant rule, so the accounting fraction is $16/18 = 88.9\%$. It is not a proportion of model variance, a population mediation estimate, or a claim that 89% of all prompt effects arise from scoring. The main paper's algebra applies because the strict and tolerant rules are applied to the same retained generations.

The four arms also give the exact contrast identity

$$(\text{BOTH} - \text{NL}) = (\text{NL} + \text{scaffold} - \text{NL}) + (\text{BOTH} - \text{NL} + \text{scaffold}). \tag{S4}$$

For Devstral the point effects are $9.7 = 2.8 + 6.9$ points, and for Gemma they are $11.1 = 1.4 + 9.7$. This partitions observed arm contrasts. It is not an estimate of separate formalism, information, and length effects. Similarly, Gemma's scaffold contrasts at wrapper, compilation, and test-success endpoints are +50.0, +8.3, and +1.4 points. These are endpoint summaries on the same tasks, not fitted causal mediation coefficients.

S7 Exploratory heterogeneity and pilot comparison

Table S12 shows the prespecified additional-specification contrast stratified after the results by task source. The overall Devstral estimate combines seven gains and one loss among Codeforces tasks with seven gains and eight losses among LeetCode tasks. Gemma has four gains and no losses in Codeforces and five gains and two losses in LeetCode. These patterns motivate a possible task-source interaction, but they do not independently confirm one. Source bundles task content, difficulty, available descriptions, and evaluation implementation.

The original pilot and reserve have the same 25% Codeforces and 75% LeetCode composition. In the pilot, Devstral's tolerant SPEC-minus-NL effects were +33.3 points in Codeforces and +38.9 in LeetCode. In the reserve they are +5.6 and −11.1. The change in the overall effect therefore cannot be attributed solely to a changed proportion of these two sources. The tasks within each group differ, so this does not identify a cause of the failed transfer.

The eleven Gemma NL development responses originally rejected for missing wrappers yield nine compilations and four additional test successes after wrapping. The original strict NL count was five, so the tolerant count

Table S12: Exploratory BOTH-minus-NL+scaffold contrasts by task source. These groups were examined after the confirmation results and are not independent confirmation of heterogeneity.

Model	Source	Tasks	Gains/losses	Effect (pp)
Devstral	Codeforces	18	7/1	+33.3
Devstral	LeetCode	54	7/8	-1.9
Gemma	Codeforces	18	4/0	+22.2
Gemma	LeetCode	54	5/2	+5.6

Table S13: Shared-arm test successes in the 24-task development pilot and 72-task reserve. Gemma NL uses tolerant scoring in both columns. Tolerant scoring was outcome-aware in development and fixed before reserved-task generation.

Model	Input	Development	Reserve
Devstral	NL	12/24	39/72
	SPEC	21/24	34/72
	BOTH	14/24	46/72
Gemma	NL	9/24	28/72
	SPEC	11/24	30/72
	BOTH	12/24	36/72

is nine. The SPEC-minus-NL gain falls from 25.0 to 8.3 points in that development sample. The comparable reserve reduction is from 25.0 to 2.8 points. The latter is the accounting result used in the main paper.

Devstral's development SPEC count of 21/24 falls to 34/72 in the reserve. Shared-arm prompt templates, checkpoint identifiers, and the recorded Ollama version match across phases. A `spec fn` appears in 23/24 development specifications and 70/72 reserve specifications. A `decreases` clause appears in 15/24 and 55/72. A mechanical tabular proxy, defined as at least five explicit branch or match-arm markers, occurs in 3/24 and 9/72. These simple descriptors do not identify all relevant information differences. They should not be used to claim that the two samples are otherwise equivalent.

S8 Numerical audit and reproduction scope

The manuscript source package contains editable LaTeX, a bibliography, generated tables, the complete 576-cell programming outcome grid, and twenty paired benchmark summaries for each predictive setting. Running `analysis/audit_results.py` reconstructs the benchmark-cluster intervals, exact programming tests, wrapper accounting, and source-stratified counts. The companion script regenerates the supporting tables. Neither script makes model requests.

The separate retained-analysis archive supplies the complete item-level out-of-fold prediction files, generation metadata and evaluation records for all 576 confirmation cells, protocols, original inference and evaluation code, and task manifests. Response-only records for all 576 confirmation cells are included, together with an extraction verifier and a rerun entry point for the 36 wrapped Gemma NL responses. The original combined prompt/response records remain excluded; their anonymised paths, byte counts, and SHA-256 hashes are retained in an omission manifest. The archive also includes the original developmental deduplication audit, its required pilot inputs, and a portable reproduction entry point. Model weights, third-party datasets and test files, benchmark problem text, and the embedding cache are not redistributed. Pinned upstream revisions and build identifiers specify those inputs.

The small audit verifies calculations from retained outcomes and paired prediction summaries. Item-level verification uses the companion probability files. A full rerun additionally requires the upstream data, tests, and exact model builds. Neither numerical recalculation nor a matching file hash independently establishes annotation validity, generated-program correctness beyond the test suite, or repeat-generation reliability.

The reproduction manifest records source roles and SHA256 values without author-specific paths. The original environment used Python 3.9.6, NumPy 1.24.4, pandas 1.5.3, SciPy 1.10.1, scikit-learn 1.6.1, and joblib 1.5.3. Direct representation contrasts and task-source splits retain their post-result status. No primary comparison is omitted or replaced by a secondary result.

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